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A method to determine stratification efficiency of thermal energy storage processes independently from storage heat losses

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Abstract

A new method for the calculation of a stratification efficiency of thermal energy storages based on the second law of thermodynamics is presented. The biasing influence of heat losses is studied theoretically and experimentally. Theoretically, it does not make a difference if the stratification efficiency is calculated based on entropy balances or based on exergy balances. In practice, however, exergy balances are less affected by measurement uncertainties, whereas entropy balances can not be recommended if measurement uncertainties are not corrected in a way that the energy balance of the storage process is in agreement with the first law of thermodynamics. A comparison of the stratification efficiencies obtained from experimental results of charging, standby, and discharging processes gives meaningful insights into the different mixing behaviors of a storage tank that is charged and discharged directly, and a tank-in-tank system whose outer tank is charged and the inner tank is discharged thereafter. The new method has a great potential for the comparison of the stratification efficiencies of thermal energy storages and storage components such as stratifying devices.
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Keywords: Thermal stratification; Thermal energy storage; Solar heating systems; Thermocline

1. Introduction

Temperature stratification in a thermal energy storage (TES) of a solar heating system may considerably increase system performance, especially for low flow solar heating systems (e.g. Lavan and Thompson, 1977; Phillips and Dave, 1982; Hollands and Lightstone, 1989; Cristofari et al., 2003; Andersen and Furbo, 2007). For the development of TES components and processes, as well as for the comparison of different TES on the market, it is desirable to have an index or "measure" to determine the ability of a TES to promote and maintain stratification during charging, storing and discharging. Several indices and numbers have been proposed for this purpose (Davidson et al., 1994; van Berkel, 1997; Shah and Furbo, 2003; Huhn, 2007; Panthalookaran et al., 2007; Andersen et al., 2007). A summary of such numbers has been given by Zurigat and Ghajar (2002), Panthalookaran et al. (2007) and Haller et al. (2009). Destratification, or mixing of two fluids with different temperatures, always results in generation of entropy and loss of exergy (Rosen et al., 2004). The better a TES process is stratifying and maintaining stratification, the less entropy is generated. Therefore, entropy generation or exergy loss can be used as a measure for the ability of a TES to promote and maintain stratification (van Berkel et al., 1999; Shah and Furbo, 2003; Rosen et al., 2004; Panthalookaran et al., 2007; Huhn, 2007; Lohse et al.,

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Nomenclature

a	time constant (s^{-1})
c	specific heat capacity (J/kg K)
C	thermal capacitance (J/K)
$\dot{C}$	capacity flow rate (W/K)
h	specific enthalpy (J/kg)
H	enthalpy (J)
i	index of summation for mass series (–)
j	index of summation for time series (–)
k	coverage factor (–)
m	mass (kg)
$\dot{m}$	mass flow (kg/s)
M	number of assumed homogenous mass elements of the TES, upper bound of summation for mass series (–)
N	upper bound of summation for time series (–)
Q	thermal energy (J)
$\dot{Q}$	thermal power (W)
s	specific entropy (J/kg K)
S	entropy (J/K)
$\dot{S}$	entropy flow rate (W/K)
T	absolute temperature (K)
τ	time (s)
$\Delta\tau$	time step (s)
UA	overall heat transfer coefficient area product for heat losses to ambient (W/K)
η	stratification efficiency (–)
ξ	exergy (J)

Superscripts

exp	experimentally determined "real case" (generally includes heat losses and mixing)
hl	calculated including heat losses, assuming the same heat loss coefficients as for the experimental storage
mix	perfectly mixed from the beginning of the experiment by constantly mixing the inlet fluid with the entire tank fluid; the energy content is thus not the same as in the experimental storage

Subscripts

A	at the beginning of a time period with constant inlet flow rate and temperature
B	at the end of a time period with constant inlet flow rate and temperature
0	thermodynamic dead state (=state with no exergy) in the case of T_0, or start of the experiment in case of τ_0
amb	ambient of the TES
corr	correction/corrected value
flow	attributed to the charging/discharging mass flows
hl	attributed to heat losses
inf	after an infinite time
int	caused by internal processes of mixing or heat conduction
inlet	fluid entering the TES
irr	irreversible part of entropy or exergy change
ms	measured
out	fluid leaving the TES
st	stratification efficiency according to the new method presented in this article
*st*0	stratification efficiency with exergy/entropy changes caused by heat losses erroneously attributed to internal processes
store	value attributed to the TES
S	based on entropy balance
wat	water
ξ	based on exergy balance

2008). In a real TES process, exergy is not only lost by mixing or destratification, but also by heat losses. Since destratification may be viewed as a process occurring only inside of a TES, exergy losses caused by destratification have been referred to as internal exergy losses, or losses due to internal irreversibilities. Exergy losses caused by heat losses to the ambient outside the TES have been referred to as external exergy losses (Rosen et al., 1999; Rosen, 2001; Panthalookaran et al., 2007; Bakan et al., 2008).

Generally, it is of great interest how far from the ideal case of perfect stratification or the worst case of full mixing a TES process is. Therefore, stratification efficiencies or degrees of mixing that range from 0% to 100% have been defined by several authors (e.g. Davidson et al., 1994; Shah and Furbo, 2003; Panthalookaran et al., 2007; Huhn, 2007; Andersen et al., 2007). Haller et al. (2009) applied these methods to hypothetical storage processes of charging, discharging and storing. The result has been compared to the rate of entropy generation caused by internal mixing. It has been concluded that from the methods applied according to their description in literature, only the relative entropy generation as described by Huhn (2007) was always in qualitative agreement with the internal rate of entropy generation. In the hypothetical storage processes that were analyzed, heat losses have been set to zero such that all entropy generation was internal entropy generation. In a real storage process, however, heat losses can not be eliminated. Unlike other methods (e.g. Davidson et al., 1994; Andersen et al., 2007; Panthalookaran et al., 2007) the method of Huhn does not separate internal entropy generation from entropy changes due to heat losses to the ambient. Therefore, different stratification efficiencies may be obtained for two TES with different insulation, even if the stratification capability is the same for both

stores. In this article, the influence of heat losses on stratification efficiency determined by using the relative entropy generation/exergy loss methods is studied, and an attempt is made to further develop the method of Huhn in order to separate internal exergy losses from exergy losses to the ambient, or internal entropy generation from entropy changes caused by heat losses, respectively. This new method is applied to results from simulations and measurements likewise.

2. Definition of a stratification efficiency based on relative internal exergy losses

Entropy and exergy balances have been widely used in TES analysis (e.g. Rosen, 1992; Rosen et al., 1999, 2004; van Berkel et al., 1999; Shah and Furbo, 2003; Huhn, 2007; Panthalookaran et al., 2007). Unlike energy, the entropy in a closed system is not constant. In a TES process entropy may be generated by mixing as well as by heat conduction and diffusion. Thus, entropy generation is an unwanted but unavoidable phenomenon. In the following it is assumed that the border of a TES is the boundary of the system that is to be analyzed. As shown in Fig. 1, the entropy change of a TES system may be caused by mass transfer across the system boundaries as indicated by $\dot{S}_{inlet}$ and $\dot{S}_{out}$ respectively, by heat transfer across the system boundary $\dot{S}_{hl,store}$ or by internal entropy generation $\dot{S}_{irr,int}$.

The entropy change rate of a TES process $dS_{store}/d\tau$, assuming heat losses and outlet mass flows to be represented by negative values of $\dot{S}_{hl,store}$ and $\dot{S}_{out}$, is

$$dS_{store}/d\tau = \dot{S}_{inlet} + \dot{S}_{out} + \dot{S}_{hl,store} + \dot{S}_{irr,int} \tag{1}$$

For a given storage process that takes place between time 0 and time 1, the entropy change of the TES between time 0 and time 1 may be expressed as

$$\Delta S_{store} = \Delta S_{inlet} + \Delta S_{out} + \Delta S_{hl,store} + \Delta S_{irr,int} \tag{2}$$

and the internal entropy generation of the process is

$$\Delta S_{irr,int} = \Delta S_{store} - \Delta S_{flow} - \Delta S_{hl,store} \geqslant 0 \tag{3}$$

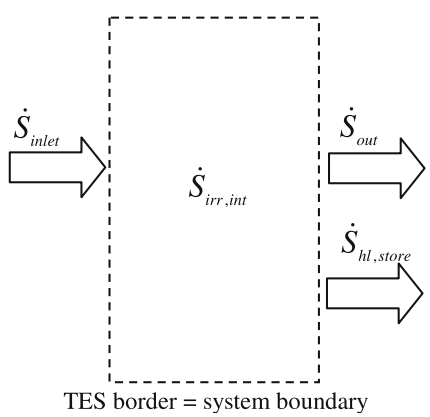

Fig. 1. System boundary for the calculation of internal entropy generation.

with

$$\Delta S_{flow} = \Delta S_{inlet} + \Delta S_{out} \tag{4}$$

The internal entropy generation may be calculated for an experimental TES ($\Delta S^{exp}_{irr,int}$), and for a fully mixed reference TES ($\Delta S^{mix,hl}_{irr,int}$). The fully mixed reference TES is assumed to be equal to the experimental TES at the beginning of the experiment, and fully mixed from this point onto the end of the experiment. Further, the fully mixed reference TES has the same heat loss coefficients as the experimental TES (indicated by the superscript *hl*), and is subjected to the same inlet mass flows and temperatures. If the experiment does not start with a uniform temperature of the experimental TES, the internal entropy generation of the fully mixed reference TES includes also the entropy generation obtained from mixing volumes with different temperatures at the beginning of the experiment. Based on these assumptions, the stratification efficiency $\eta_{st,S}$ is calculated based on the internal entropy generation of the experimental TES relative to the internal entropy generation of the fully mixed reference TES as shown in Eq. (5). This value will be 0 if the storage tank was always fully mixed during the experiment and 1 for a perfectly stratified, isentropic storage process

$$\eta_{st,S} = 1 - \frac{\Delta S^{exp}_{irr,int}}{\Delta S^{mix,hl}_{irr,int}} \tag{5}$$

Another route for the calculation of this stratification efficiency is based on the internal exergy loss of the experimental TES $\Delta\xi^{exp}_{irr,int}$ relative to the internal exergy loss of a fully mixed reference TES $\Delta\xi^{mix,hl}_{irr,int}$, i.e.:

$$\eta_{st,\xi} = 1 - \frac{\Delta\xi^{exp}_{irr,int}}{\Delta\xi^{mix,hl}_{irr,int}} \tag{6}$$

With $\Delta\xi_{hl,store}$ being a negative value in the case of net heat losses, $\Delta\xi_{irr,int}$ is defined as

$$\Delta\xi_{irr,int} = \Delta\xi_{store} - \Delta\xi_{flow} - \Delta\xi_{hl,store} \leqslant 0 \tag{7}$$

In this case, the exergy changes are calculated based on the enthalpy changes ΔH, the entropy changes ΔS, and the thermodynamic dead-state temperature T_0

$$\Delta\xi = \Delta H - T_0 \cdot \Delta S \tag{8}$$

Because internal mixing does not change the enthalpy content of the TES ($\Delta H_{int} = 0$), the internal exergy loss of a TES is directly proportional to the internal entropy generation. Thus, the stratification efficiency based on the relative internal exergy loss equals the stratification efficiency based on relative internal entropy generation

$$T_0 \cdot \Delta S_{irr,int} = -\Delta\xi_{irr,int} \tag{9}$$

$$\eta_{st,\xi} = \eta_{st,S} \tag{10}$$

This identity has already been pointed out by Huhn (2007).

The enthalpy and entropy change of the TES caused by charging and discharging processes, assuming that the inlet

mass flow equals the outlet mass flow, evaluated at the time τ_1, are:

$$\Delta H_{flow}(\tau_1) = \int_{\tau_0}^{\tau_1} \dot{m}(\tau) \cdot [h\{T_{inlet}(\tau)\} - h\{T_{out}(\tau)\}] \cdot d\tau \quad (11)$$

$$\Delta S_{flow}(\tau_1) = \int_{\tau_0}^{\tau_1} \dot{m}(\tau) \cdot [s\{T_{inlet}(\tau)\} - s\{T_{out}(\tau)\}] \cdot d\tau \quad (12)$$

The enthalpy and entropy change of the TES, from the beginning of the experiment until the time τ_1 are:

$$\Delta H_{store}(\tau_1) = \int_0^{m_{store}} [h\{T_{store}(\tau_1, m)\} - h\{T_{store}(\tau_0, m)\}] dm \quad (13)$$

$$\Delta S_{store}(\tau_1) = \int_0^{m_{store}} [s\{T_{store}(\tau_1, m)\} - s\{T_{store}(\tau_0, m)\}] dm \quad (14)$$

The enthalpy change due to heat losses is calculated based on the heat transfer coefficient area products $UA(m)$ between each mass element and the ambient:

$$\Delta H_{hl,store}(\tau_1) = \int_{\tau_0}^{\tau_1} \int_0^{m_{store}} \frac{-UA(m) \cdot \{T_{store}(\tau, m) - T_{amb}(\tau)\}}{m_{store}} dm \cdot d\tau \quad (15)$$

With $\dot{S} = \dot{Q}/T$ given for the entropy change rate associated with heat transfer, the entropy change of the TES caused by heat losses is:

$$\Delta S_{hl,store}(\tau_1) = \int_{\tau_0}^{\tau_1} \int_0^{m_{store}} \frac{-UA(m) \cdot \{T_{store}(\tau, m) - T_{amb}(\tau)\}}{m_{store} \cdot T_{store}(\tau, m)} dm \cdot d\tau \quad (16)$$

For the calculation of the fully mixed reference store, the time dependent inlet temperatures and mass flows and the ambient temperatures are identical to the ones that are measured (or simulated) for the experimental TES. The TES temperature of the fully mixed reference TES is at the same time its outlet temperature. For time periods with constant charging or discharging mass flow and inlet temperature, the temperature of the fully mixed reference TES at time τ_B is calculated from the temperature at the beginning of this period $T_{store}^{mix,hl}(\tau_A)$, the temperature T_{inf} that would be reached after an infinite time, and the time constant a of the process:

$$T_{out}^{mix,hl}(\tau_B) = T_{store}^{mix,hl}(\tau_B) = T_{inf} + [T_{store}^{mix,hl}(\tau_A) - T_{inf}] \cdot exp(-a \cdot [\tau_B - \tau_A]) \quad (17)$$

with

$$T_{\inf} = \frac{UA \cdot T_{amb} + \dot{C}_{flow} \cdot T_{inlet}}{UA + \dot{C}_{flow}} \quad (18)$$

$$a = \frac{UA + \dot{C}_{flow}}{C_{store}} \quad (19)$$

where $\dot{C}_{flow}$ is the capacity flow rate of the charging or discharging process, and C_{store} is the thermal capacitance of the TES

$$\dot{C}_{flow} = \dot{m}_{flow} \frac{h(T_{out}) - h(T_{inlet})}{T_{out} - T_{inlet}} \quad (20)$$

3. Application to a simulated experiment

3.1. Description of the simulation

Simulations of the charging and discharging experiment described in Table 1 were performed with a simplified 10 nodes TES model programmed in Visual Basic. The simulation time step of 0.1 h corresponded to the time needed to replace the fluid in one node by the fluid from the upstream node during the charging and discharging processes. This way, numerical diffusion was avoided. The only sources of internal entropy generation for the simulated experimental TES were therefore caused by the well defined processes of effective vertical thermal conductivity and mixing listed in Table 2.

In the model, the temperatures of all nodes were calculated in a stepwise manner according to the following sequence, without the need of iterations:

1. Fluid replacement during the charging or discharging process.
2. Mixing of three nodes in the inlet region of the charging or discharging process for the simulation of inlet jet mixing.
3. Vertical diffusion and conduction of heat based on the temperature distribution resulting from the previous calculation steps.
4. Heat losses to the ambient based on the difference between the ambient temperature and the average temperature of each node at the beginning and at the end of this time step (using the results of step 3).

The calculation of the fully mixed reference TES was performed according to Eqs. (17)–(20) and Table 1.

Stratification efficiencies based on relative entropy generation $\eta_{st,S}$ and relative exergy losses $\eta_{st,\xi}$ were calculated using Eqs. (3)–(8). These numbers are compared to results based on a calculation that does not account for exergy and

Table 1
Assumptions for the simulated charging and discharging experiments.

Mass of the TES	140 kg
Fluid/material specific heat	4.18 kJ/kgK
Initial temperature	20 °C
Mass flow for charging and discharging	140 kg/h
Inlet position for charging = outlet position for discharging	Top
Inlet position for discharging = outlet position for charging	Bottom
Charging time period[a]	Hours 1–1.8
Charging temperature	50 °C
Standby time period[a]	Hours 1.8–6
Discharging time period[a]	Hours 6–10
Discharging temperature	20 °C
Overall heat loss coefficient of simulation 1[b]	0.5 W/K
Overall heat loss coefficient of simulation 2[b]	2.0 W/K

[a] From the beginning of the experiment.
[b] Distributed evenly between all mass elements.

Table 2
Assumptions for mixing within the simulated experimental TES.

Number of nodes	10
Number of mixed nodes at top (during charging)	3
Number of mixed nodes at bottom (during discharging)	3
Height	1 m
Cross-sectional area	0.14 m^2
Fluid/material density	1000 kg/m^3
Effective vertical thermal conductivity	2.5 W/mK

entropy changes due to heat losses in both, the experimental TES and the fully mixed reference TES, as given by Eqs. (21) and (22)

$$\eta_{st0,S} = 1 - \frac{\Delta S^{exp}_{store} - \Delta S^{exp}_{flow}}{\Delta S^{mix}_{store} - \Delta S^{mix}_{flow}} \tag{21}$$

$$\eta_{st0,\xi} = 1 - \frac{\Delta \xi^{exp}_{flow} - \Delta \xi^{exp}_{store}}{\Delta \xi^{mix}_{flow} - \Delta \xi^{mix}_{store}} \tag{22}$$

For the calculation of exergy, a thermodynamic dead-state temperature of $T_0 = 298.15$ K has been assumed.

3.2. Results and discussion

Fig. 2a shows that the stratification efficiencies $\eta_{st0,S}$ and $\eta_{st0,\xi}$ that were calculated without taking into account the entropy/exergy change caused by heat losses are not equal to each other ($\eta_{st0,S} \neq \eta_{st0,\xi}$). The difference is quite large, even though a relatively small heat loss coefficient of 0.5 W/K has been assumed for the TES of 140 kg water. If the entropy/exergy change due to heat losses is separately accounted for, the stratification efficiency based on internal entropy generation is, as expected, equal to the stratification efficiency based on internal exergy losses ($\eta_{st,S} = \eta_{st,\xi}$). From the comparison of Fig. 2a and b two main conclusions can be drawn. First, the deviation of the calculated stratification efficiencies $\eta_{st0,S}$ and $\eta_{st0,\xi}$ becomes larger with increasing heat loss coefficients of the TES. Second, no notable difference can be seen between the $\eta_{st,S}$ of the two example calculations with different heat loss coefficients.

Neglecting the influence of heat losses on the exergy balance of the TES obviously results in reduced values for the stratification efficiency based on the exergy balance. This can be explained by the fact that exergy loss associated with heat losses are unintentionally added to the internal exergy loss in the calculation procedure, thereby increasing this value and decreasing the calculated stratification efficiency. At first sight it might come as a surprise that if the stratification efficiency is calculated based on the entropy balance, neglecting the effect of heat losses has the opposite effect. This is a consequence of the entropy flow associated with heat transfer. If a TES is loosing energy to the ambient it looses at the same time entropy. Therefore, not accounting for this entropy loss separately will decrease the calculated internal entropy generation.

4. Application to experimental results

4.1. Measurement setup and uncertainties

Measurements were performed on two solar TES that may be used for the storage of thermal energy for domestic hot water preparation and space heating. TES1 is an 800 l storage tank with an integrated biomass burner that has been tested in direct charging and discharging mode (Fig. 3, left). It contained internal heat exchangers for solar charging and domestic hot water production that were filled with water during the experiments. The integrated burner and the internal heat exchangers have not been used during the experiments and are therefore not shown in Fig. 3. TES2 is a tank-in-tank storage system (Fig. 3, right) with a total volume of 124.5 l, of which 29.5 l were contained in the inner tank.

Tables 3 and 4 show the applied measurement instruments and their estimated uncertainties for TES1 and TES2, respectively. Inlet and outlet flow rates of TES1 were measured with a magnetic inductive sensor that has been calibrated with a scale and a stop-watch in the range of 100–900 l/h. Inlet and outlet temperatures were measured with immersed 4 wire Pt100 class A sensors that have been calibrated simultaneously against another 4 wire Pt100 class A sensor in a thermostat bath in the range of 10–90 °C. A

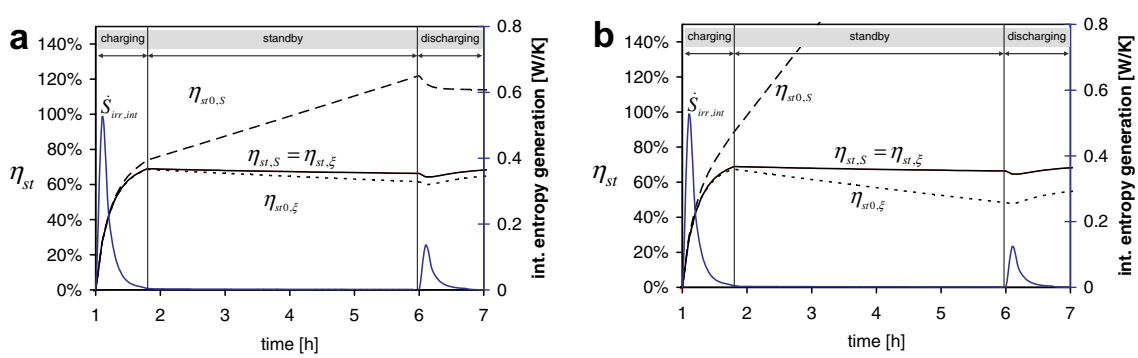

Fig. 2. Second law stratification efficiencies for a simulated TES process with heat losses of 0.5 W/K (a) and 2.0 W/K (b).

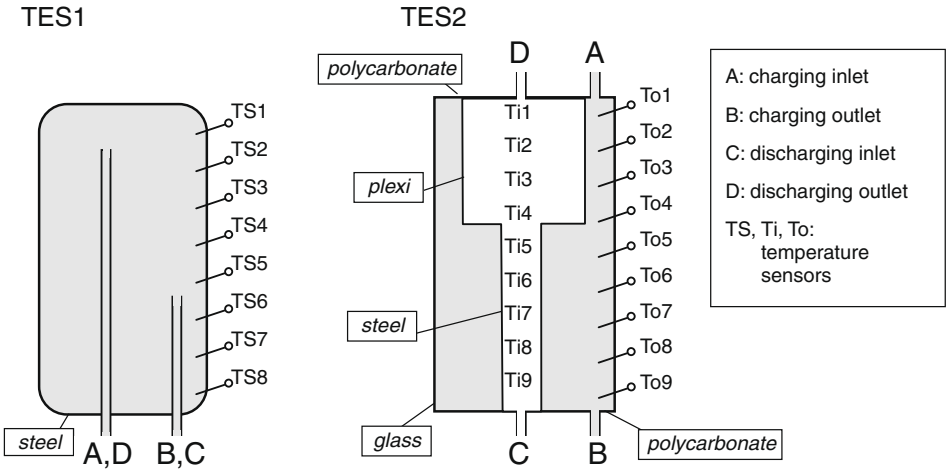

Fig. 3. Simplified schematic pictures of the two TES that have been used for experiments.

total of eight thermocouple sensors of Type K were attached to the outside of the steel tank, below its insulation, equally spaced along a vertical line. Then, they were calibrated against the average of the inlet and outlet temperatures of the flow and return line in steady state in the range of 20–60 °C. Measurements were taken at intervals of 1 s and averaged values were stored at intervals of 10 s.

TES2 was measured in a different laboratory, and therefore different measurement devices have been used. Inlet and outlet flow rates of TES2 were measured with vortex flow sensors (VFS) that have been calibrated with a scale and a stop-watch in the range of 120–840 l/h. Inlet and outlet temperatures were measured with immersed copper/constantan thermopiles whose uncertainty is estimated according to Furbo (1984). A total of 18 thermocouple sensors of Type TT were immersed into the TES, nine of them in the outer tank and nine of them in the inner tank, as shown in Fig. 3, all of them with known positions on a vertical line. Measurements were taken at intervals of 5 s and stored at intervals of 5 s.

Table 3
Measurement devices and uncertainty used for the measurements on TES1.

Measured property	Device	Uncertainty (k = 1)
Volume flow rate	Endress and Hauser Promag P50	±0.6%
Temperature (flow)	4w Pt100 class A	±0.1 K
Temperature difference (flow inlet and outlet)	4w Pt100 class A	±0.03 K
Temperature (tank)	Type K thermocouple	±0.5 K

Table 4
Measurement devices and uncertainty used for the measurements on TES2.

Measured property	Device	Uncertainty (k = 1)
Volume flow rate charging	Vortex flow sensor (VFS)	±7.6%
Volume flow rate discharging	Vortex flow sensor (VFS)	±7.1%
Temperature (flow)	Copper/constantan thermocouple	±0.5 K
Temperature difference (flow inlet and outlet)	Copper/constantan thermopiles	±0.05 K
Temperature (tank)	Type TT thermocouple	±0.5 K

4.2. Calculation procedure

A number of discrete time steps $j = 1 \ldots N$ were used instead of a continuous integration over time for the calculation of enthalpy and entropy changes due to mass flows and heat losses. The calculation time steps were $\Delta\tau = 10$ s for TES1, and $\Delta\tau = 5$ s for TES2. A number of discrete mass elements ($M = 8$ for TES1 and $M = 18$ for TES2) that correspond to the number of sensors used to measure the temperatures inside the TES have been used for the calculation of enthalpy changes of the TES, together with a thermal capacitance $C_{store,i}$ associated with each sensor measurement i.e.:

$$\Delta H_{store} = \sum_{i=1}^{M} C_{store,i} \cdot [T_{store,i}(\tau) - T_{store,i}(\tau_0)] \tag{23}$$

The total thermal capacitance C_{store} of TES1 has been measured with a complete charge and discharge test, and the mass of each TES element corresponding to a different temperature sensor has been calculated with $C_{store,i} = C_{store}/M$. Because of the different volumes associated with the different temperature sensors of TES2, $C_{store,i}$ has been calculated for each sensor individually based on the geometrical data of TES2. The thermal capacitance of the outer wall, without insulation, has been attributed to the corresponding fluid mass element and temperature inside the TES. The thermal capacitance of the wall separating the inner tank from the outer tank has been attributed half to the mass element of the inner tank and half

to the mass element of the outer tank. The properties of the materials used for this calculation are shown in Table 5.

To calculate the entropy change of the TES, an equivalent water mass $m_{store,i} = C_{store,i}/c_{wat}$ has been calculated for each mass element with $c_{wat} = 4.18$ kJ/kg K. The entropy change was then calculated using the entropy function of water, i.e.:

$$\Delta S_{store} = \sum_{i=1}^{M} m_{store,i} \cdot [s\{T_{store,i}(\tau)\} - s\{T_{store,i}(\tau_0)\}] \tag{24}$$

The enthalpy change and entropy change caused by heat losses were calculated with Eqs. (25) and (26):

$$\Delta H_{hl,store} = \sum_{j=1}^{N} \sum_{i=1}^{M} -UA_i \cdot \{T_{store,i}(j) - T_{amb}(j)\} \cdot \Delta\tau \tag{25}$$

$$\Delta S_{hl,store} = \sum_{j=1}^{N} \sum_{i=1}^{M} \frac{-UA_i \cdot \{T_{store,i}(j) - T_{amb}(j)\}}{T_{store,i}(j)} \cdot \Delta\tau \tag{26}$$

For the calculation of heat losses for the single mass elements, overall heat transfer coefficient area products UA_i have been obtained by simulating the cooling out behavior during a standby period, and comparing the simulation results with the measurements. For TES1, which was not evenly insulated, the TES was split into three different zones with different overall heat loss coefficients. TES2 was evenly insulated and therefore a constant heat loss coefficient has been assumed for all surface areas.

Measurement uncertainties inevitably lead to calculated results for $\Delta H_{store,ms}$, ΔH_{flow}, and $\Delta H_{hl,store}$ that are not in perfect agreement with the law of conservation of energy ($\Delta H_{store,ms} \neq \Delta H_{flow} + \Delta H_{hl,store}$). It is reasonable to assume that the first law of thermodynamics will never be violated, and thus it can be justified that the measured results are "corrected" within the limits of measurement uncertainties in order to be in agreement with the first law of thermodynamics. Inevitably, the question arises which measured values should be corrected. For the calculation of the energy balance, each TES temperature sensor has been assumed to represent the temperature of a certain section/mass element of the TES. In reality, the temperatures within these sections or mass elements are not uniform. This adds additional uncertainty to the uncertainty of the temperature measurement at the location of the sensor itself. Additionally, the temperature sensors attached to the outer surface of TES1 register temperature changes inside the TES only with a time delay, thus contributing further to the uncertainty of values calculated based on these temperatures. For this reason, the mentioned correction has been applied to the measured TES temperature measurements, i.e.:

$$T_{store,i}(j) = T_{store,i,ms}(j) + \Delta T_{store,corr}(j) \tag{27}$$

The correction term $\Delta T_{store,corr}(j)$ is chosen within the uncertainties of the energy balance, such that at each time step:

$$\Delta H_{store} = \Delta H_{flow} + \Delta H_{hl,store} \tag{28}$$

For the calculation of exergy, a thermodynamic dead-state temperature of $T_0 = 298.15$ K has been assumed.

4.3. Description of the experiments

The experiments that have been performed with the two TES shown in Fig. 3 consisted of three sequential periods. The first period was a heat charging process that started from a uniform TES temperature and ended when about 75% of the TES volume had been replaced. This was followed by a standby period (i.e., a period without fluid exchange) of about 4 h. In the third and last period, the stored energy was discharged, if possible until the initial temperature was reached again. All inlet temperatures and mass flows were kept constant during the charging and discharging processes respectively.

Four experiments with different mass flows and different insulation were performed with TES1. Nominal values of these experiments are shown in Table 6. Different flow rates were chosen in order to evaluate their influence on the stratification efficiency. Different insulation was chosen in order to determine the ability of the new method presented in Section 2 to reduce the biasing influence of heat losses on the calculated stratification efficiency.

4.4. Results and discussion

Fig. 4a shows the measured temperatures (TS1–TS8) of TES1 over the time of experiment 2 listed in Table 6. The simulated temperature of the fully mixed reference store $T_{store}^{mix,hl}$ is also shown.

Table 5
Materials and properties for the calculation of the thermal capacitance of TES2.

Material	Volume (m^3)	Specific heat (kJ/m^3 K)
Water	0.1216	4148
Glass	0.0164	1890
Polycarbonate	0.0017	1404
Plexiglass	0.0014	1786
Steel	0.0013	3398

Table 6
Nominal values for the different experiments performed on TES1.

Experiment		1	2	3	4
Overall heat loss coefficient	W/K	11	11	17	17
Charging/discharging mass flow rate	kg/h	200	400	200	400
Charging time	h	3.0	1.5	3.0	1.5
Standby time	h	4.0	4.0	4.0	4.0
Discharging time	h	6.0	4.5	6.0	4.5
End of the experiment	h	13	10	13	10
Initial tank temperature[a]	°C	20	20	20	20
Charging inlet temperature	°C	60	60	60	60

[a] Initial tank temperature is at the same time discharging inlet temperature and temperature at the end of the experiment.

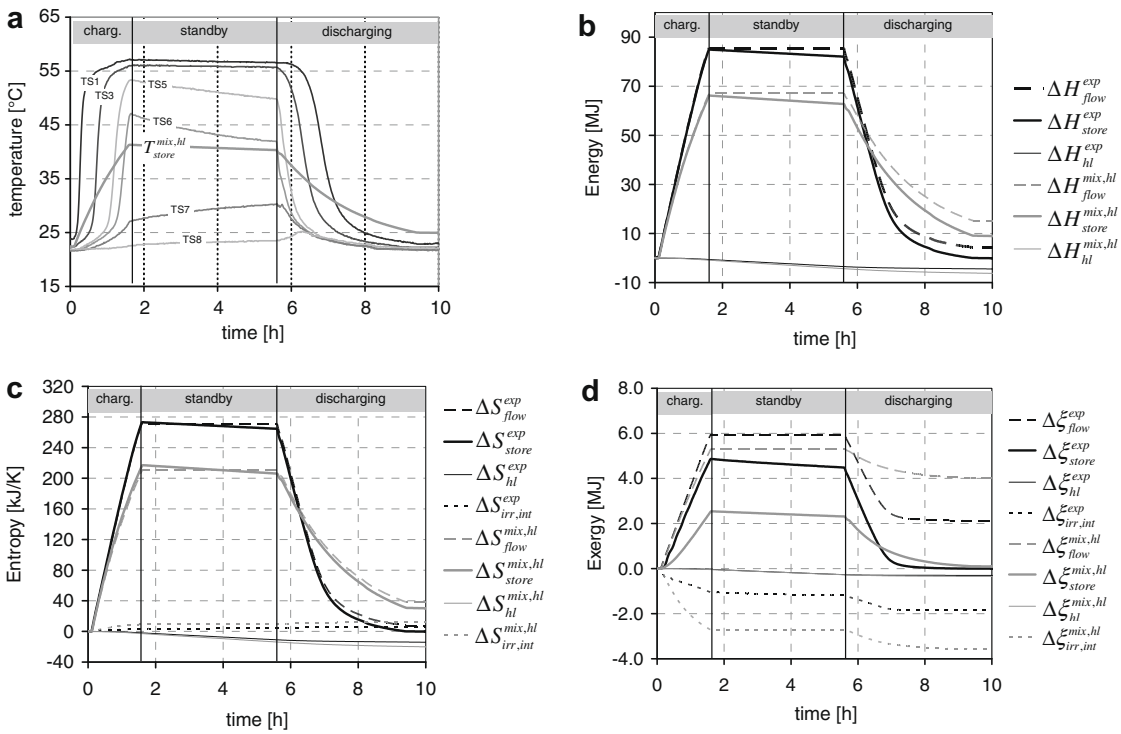

Fig. 4. Temperature profiles (a), enthalpy changes (b), entropy changes (c), and exergy changes (d) during the charging, standby, and discharging sequence of TES1.

Figs. 4b–d show the calculated enthalpy changes, entropy changes, and exergy changes of the same experiment. If the total turnover of enthalpy, entropy, and exergy is defined as the maximum values of ΔH_{flow}^{exp}, ΔS_{flow}^{exp}, $\Delta \xi_{flow}^{exp}$, respectively, at the end of the charging process, then the figures show that the changes due to heat losses that are represented by ΔH_{hl}^{exp}, ΔS_{hl}^{exp}, and $\Delta \xi_{hl}^{exp}$ are only a small fraction of the respective total enthalpy, entropy and exergy turnover. The internal entropy generation $\Delta S_{irr,int}^{exp}$ shown in Fig. 4c is small compared to the total entropy turnover of the experiment. It is even smaller – in absolute values – than the entropy loss caused by heat losses ΔS_{hl}^{exp}. This underlines the importance of a proper estimation of heat losses when working with entropy balances. However, the exergy loss caused by internal irreversibilities $\Delta \xi_{irr,int}^{exp}$ shown in Fig. 4d is much larger than the exergy loss caused by heat losses $\Delta \xi_{hl}^{exp}$.

Fig. 5a shows the calculated stratification efficiency over the time of the experiment. The stratification efficiency curve shows similar characteristics as the one in the simulated experiment shown in Fig. 2. The wavy shape of the curve during the charging process is interpreted as a bias caused by the limited number of temperature sensors of the TES. The frequency of the waves corresponds well to the time lag between the temperature response of one sensor and the next sensor in the path of the fluid moving

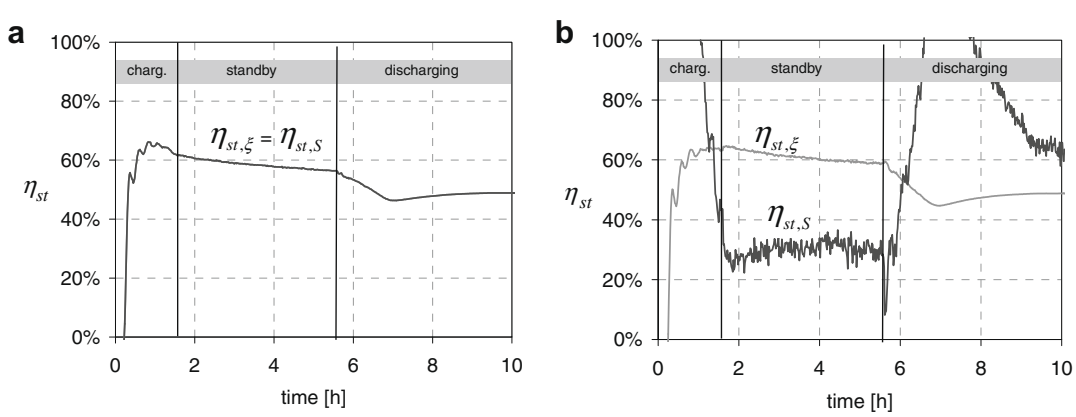

Fig. 5. Stratification efficiency during the charging, standby, and discharging sequence of TES1 with correction (a) and without correction (b) of TES temperatures.

inside the TES (see Fig. 4a). For the interpretation of the results, it has to be kept in mind that the stratification efficiency is an integrated value over the whole experiment. Thus, stratification efficiency at a certain time of the experiment is not independent from stratification efficiencies and experimental conditions of previous times within the experiment.

The decrease of η_{st} during standby can be attributed to internal heat conduction and convection. The decrease of η_{st} at the beginning of the discharging process corresponds to a time where cold fluid flowing into the TES meets hotter fluid in the TES. During the discharge, the temperature at the point of the inlet decreases, and thus also the temperature difference of the mixing fluids decreases and consequently also the internal exergy loss rate. Fig. 5b shows that if uncorrected values are taken for the TES temperatures, then the stratification efficiencies based on entropy $\eta_{st,S}$ and based on exergy $\eta_{st,\xi}$, calculated with Eqs. (3) and (5), and with Eqs. (7) and (6), respectively, are not equal. In this case, the difference between $\eta_{st,S}$ and $\eta_{st,\xi}$ at the end of the experiment is 10%, although the deviation of the energy balance at the end of the experiment is only 0.5% of the total energy turnover of the experiment.

Since the stratification efficiency η_{st} is based on integrated values of entropy or exergy changes, its value at the end of an experiment may be considered to be of more importance than a value at the beginning of an experiment. Furthermore, in the experimental results shown here, the temperature of the TES at the end of the experiment is uniform, and similar to its temperature at the beginning of the experiment. Thus, uncertainties associated with the determination of $\Delta H^{exp}_{store}, \Delta S^{exp}_{store}$, and $\Delta\xi^{exp}_{store}$ are lower at the end of the experiment.

Fig. 6 shows different stratification efficiencies η_{st} calculated for TES1 at the end of the experiments listed in Table 6. For two of the experiments, heat losses have been increased from 11 W/K to 17 W/K by removing parts of the insulation. Fig. 6 shows that $\eta_{st,\xi}$ are 7–10% points higher than $\eta_{st0,\xi}$ for the same reason as in the simulation results shown in Fig. 2. At the same time, the differences between the stratification efficiencies obtained with different insulation are significantly higher for the calculations without heat loss correction $\eta_{st0,\xi}$ than for the calculations that account separately for the exergy lost to the ambient $\eta_{st,\xi}$. It is an open question if the remaining differences of 1.0–1.5% points between the stratification efficiencies $\eta_{st,\xi}$ for experiments with different insulation is a bias caused by the uncertainty of measurement and calculation, a bias of the method itself, or a result of different amounts of internal irreversibilities/mixing that occurred in reality during the performed experiments. An uncertainty of 20% for the determined heat loss coefficients e.g. leads to uncertainties of $\eta_{st,\xi}$ in the order of 1.1–1.7% points for the experiments shown.

At first sight it is surprising that the stratification efficiencies of the experiments performed with mass flow rates of 400 kg/h are higher than the ones performed with 200 kg/h. Due to the higher kinetic energy of the inlet stream, more inlet jet mixing is expected for an experiment with a higher mass flow. Fig. 7 shows that the temperature changes during the 400 kg/h charging process are indeed steeper, and the resulting temperatures at the top of the TES are higher for this experiment. This is in agreement with the higher stratification efficiency calculated for this experiment. It is possible that heat transfer from the hot inlet pipe that enters the TES at the bottom and crosses the cold tank section before it releases the fluid in the upper part of the TES is the source of more temperature degradation during the slower charging process. Since the charging time of the experiment with 200 kg/h was twice as long as the one with 400 kg/h, more heat was lost from the fluid moving up to the end of the pipe, and transferred to the bottom section of the TES. This process has been reported as a significant source of temperature degradation for solar TES by Thür (2007).

The evolvement of the temperatures during the experiment with the tank-in-tank TES2 is shown in Fig. 8.

In Fig. 9, a significant decrease in stratification efficiency can be seen just after the charging process for TES2. This decrease may be attributed to the process of heat transfer

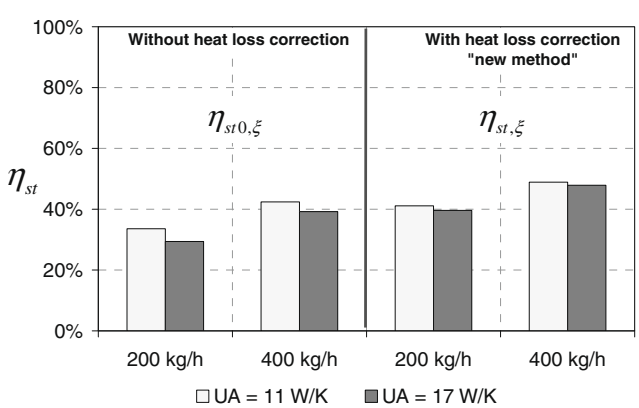

Fig. 6. Stratification efficiencies at the end of the experiments of TES1 as listed in Table 6.

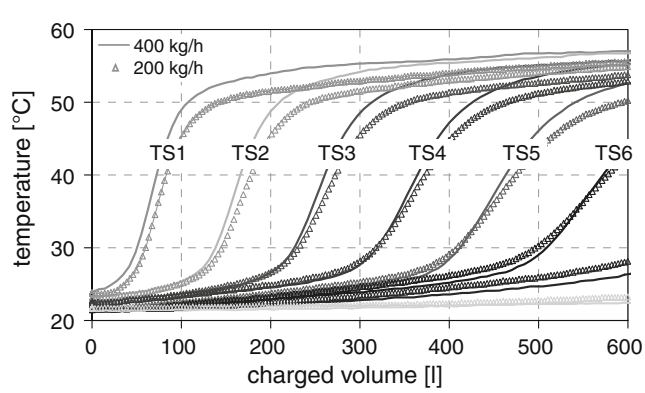

Fig. 7. Temperatures of TES1 during two charging processes with different mass flows.

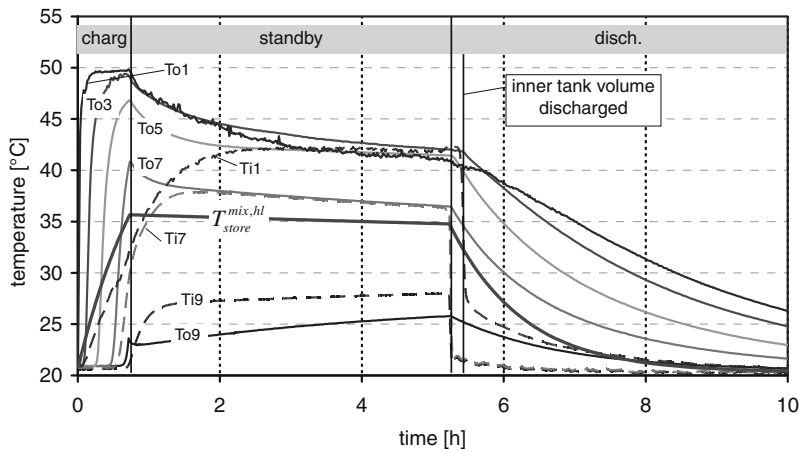

Fig. 8. Temperature profiles during the charging (charg), standby, and discharging sequence of TES2 (tank-in-tank). T_i, T_o = inside (i) and outside (o) tank temperatures (see Fig. 3).

from the outer tank of the TES to the inner tank of the TES. As a result of the second law of thermodynamics, every heat transfer process generates entropy. During the discharging of the inner tank, it can be noted that the stratification efficiency drops to values comparable to those of TES1. Only when the inner tank is almost completely discharged, the stratification efficiency drops to very low values. This can be attributed to the limited heat transfer rate from the outer part of the tank to the inner part, which leads to exergy losses comparable to those of any under dimensioned heat exchanger. This result is interesting for the evaluation of the capability of the new method, but it cannot be used to answer the question which TES concept is better from the point of view of stratification.

The experiments used to test the two TES were chosen in order to proof the concept of the stratification efficiency calculation, but they do not correspond to charging and discharging profiles that are relevant for the operation of these TES in solar thermal systems. The question inevitably arises what a relevant test profile of charging, standby and discharging for solar TES should look like for the determination of a meaningful and representative stratification efficiency. To our knowledge, this question has not been addressed systematically yet.

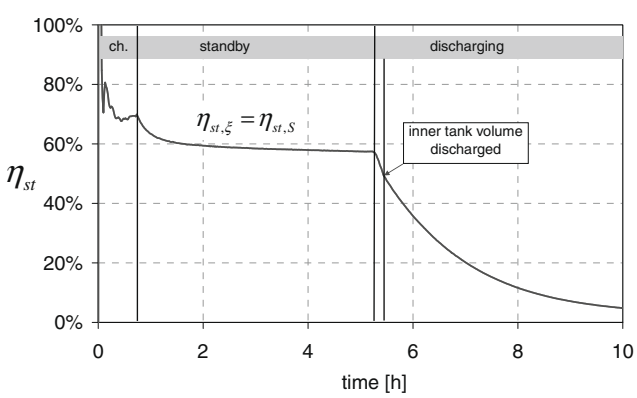

Fig. 9. Stratification efficiency during the charging (ch), standby, and discharging sequence of TES2.

5. Conclusions

A stratification efficiency based on relative exergy loss or relative entropy generation of an experimental TES has been defined. This stratification efficiency is independent from the dead-state temperature T_0 chosen for the exergy calculation. A fully mixed reference TES with equal heat loss coefficients has been used for the reference process. Theoretically, the influence of heat losses can be eliminated with this method, as has been shown with simulation results. It has also been shown by simulation and measurements that even for the case of moderate heat losses, the effect of neglecting heat losses in the calculation of stratification efficiencies based on relative exergy losses/entropy generation of the TES is significant.

The application of the new method to experimental results showed that small deviations in the energy balance that are caused by measurement uncertainties have a large effect on the entropy balance and the stratification efficiency calculated based on the entropy balance. Therefore, measured results have to be corrected within the limits of measurement uncertainties in order to be in agreement with the first law of thermodynamics. Only if this is the case, the stratification efficiency calculated based on the entropy balance is identical with the stratification efficiency based on the exergy balance. With this correction applied to experimental results, the new method was able to significantly reduce the differences between calculated stratification efficiencies of comparable TES processes performed with higher and lower insulation of the TES. However, it is recommended that the stratification experiments are done with well insulated TES since the uncertainty of the calculated stratification efficiency increases with increased uncertainty and amount of the heat lost by the TES.

The stratification efficiency curves obtained from experimental results correspond well to the curves that had been

obtained previously by simulations. The higher stratification efficiency obtained for a charging process with larger mass flow was in agreement with the better conservation of the charging temperature for the same experiment. Comparison of the stratification efficiency obtained for a storage process of a directly charged and discharged TES and a tank-in-tank TES whose outer tank was charged and whose inner tank was discharged thereafter gave meaningful insights into the different mixing processes taking place inside the two TES at different times of the experiments.

If temperatures inside the TES are known, a meaningful stratification efficiency may be calculated and plotted over the whole time of the experiment. Theoretically, the presented stratification efficiency calculation method is also suitable for varying inlet temperatures and mass flows. Stratification efficiency not only depends on the TES itself, but also on the test profile, i.e., the temperature and mass flow rate profile of the experiment. Test profiles should be developed depending on the intended application of a TES. For solar thermal storages, a relevant test profile is likely to include variable temperatures of charging and discharging. The development of such a profile is still a topic of future work. The presented method, together with relevant test profiles, has a great potential for the comparison of the stratification efficiencies of storage systems and storage components such as stratifying devices.

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